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Quantum summation using d-level entanglement swapping

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Abstract
In this paper, we propose a secure multi-party quantum modulo-d summation protocol using d-level entanglement swapping, where a malicious but non-collusive third party (TP) who plays a role of entanglement distribution between the participants is involved. TP is assumed to be malicious but non-collusive, which means TP can launch various attacks within boundaries of quantum mechanics except the collusion with any dishonest participant. Our protocol is congenitally free from Trojan horse attacks because there is no need to relay encoded states. In addition, only d-level Bell measurement is required for the users except TP, which makes our protocol's implementation more feasible. We show that the presented protocol is secure against outside as well as participant attacks. Second, we give a detailed comparison of different quantum modulo-d summation protocols, in terms of qudit cost, quantum resources, and quantum operations. In the end, we discuss how to replace cat states with d-level Bell states in Je et al.'s protocols (Quantum Inf Process 18(6):168, 2019) for easier implementation with current quantum technology.

Keywords Quantum multi-party computation · Quantum summation · Entanglement swapping

1 Introduction

Secure multi-party quantum summation [1–5], as a fundamental primitive of secure multi-party quantum computation [6–10], has been gaining attention recently. The

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goals of quantum summation are to ensure the correctness of summation result, as well as to preserve the privacy of users' inputs. Quantum summation can be applied to quantum voting [11–15], quantum anonymous ranking [16], and quantum private equality comparison [17–19]. Devising efficient and practical quantum summation protocols is thus significant, as we pursue in this work.

A variety of quantum summation protocols have been investigated by using various quantum resources. Reference [20] presented a quantum summation protocol based on single photons in both polarization and spatial-mode degrees of freedom, where unitary operations employed to encode users' private bits on the travelling single photons are required. Most other protocols depend on sharing a multipartite entangled state among users. For example, Zhang et al. [21] designed a quantum summation protocol without a trusted third party, in which the number of users was limited to three because of the use of the so-called genuinely maximally entangled six-qubit states. In 2016, Shi et al. [22] proposed protocols for summation and multiplication that involve quantum Fourier transform, controlled-NOT (CNOT) gates and oracle operators. They then studied a common quantum solution that addresses a class of two-party private summation problems [23]. In 2017, Zhang et al. [24] devised a multi-party quantum summation protocol without a trusted third party where a multipartite entangled state is shared among users. Liu et al. [25] presented a quantum summation protocol by sharing multipartite entangled states that include Bell states as subsystems among users. In their protocol, quantum communication is two-way, which means they should offer special care concerning Trojan horse attacks [26–28]. In contrast to these protocols, Zhang et al. [29] employed quantum teleportation to devise a quantum summation protocol, in which a certain entanglement distribution protocol is involved.

Recently, Ji et al. [30] discussed how to construct quantum private comparison, quantum anonymous ranking, and quantum anonymous voting protocols based on quantum summation. They designed two quantum summation protocols by employing entanglement swapping property between the d-level Bell state and the d-level n-partite cat state. As the d-level n-partite cat state remains difficult to prepare with the current technology, we try to replace them with d-level Bell states.

In some quantum cryptographical protocols, a semi-honest third party (TP) is usually involved to help compute some results [5,20]. TP is said to be semi-honest if TP executes the protocols loyally, writes down a record of all its intermediate computations, and might try to steal the participants' private inputs from the record, but he cannot collude with other participants. In our protocol, a malicious but non-collusive TP is introduced to help the computation. TP is said to be malicious but non-collusive if he can do whatever he wants within boundaries of quantum mechanics except the collusion with dishonest participants. The difference between the semi-honest TP and the TP in our protocol is that the TP in our protocol does not need to execute the protocol loyally; that is to say, the malicious TP has more possible ways to attack the protocol, which in return makes the protocol more secure. For this point, we give the assumption of the malicious but non-collusive TP.

In this work, we start with a quantum modulo-d protocol relying on the entanglement swapping property between d-level Bell states in which the malicious but non-collusive TP is introduced. A security analysis shows that our protocol can resist both outside and participant attacks. We then compare in detail our protocol with exist-

ing quantum modulo-d summation protocols. Finally, we show that the cat states in Je et al.'s protocols can be replaced with d-level Bell states, which facilitates the protocol implementation. Note that if the TP in our protocol could launch collusion attacks with some dishonest participant, they can learn some honest participant's private input. We will consider such attacks in the future and try to minimize the security assumptions.

The rest of this paper is organized as follows. In Sect. 2, we introduce the entanglement swapping property of d-level Bell states. In Sect. 3, we describe the quantum modulo-d summation protocol, followed by its correctness and security analysis in Sect. 4. Section 5 gives a detailed comparison of quantum modulo-d summation protocols and a discussion on how to replace cat states with d-level Bell states in Ji et al.'s protocols as well. Conclusions are given in Sect. 6.

2 Preliminary knowledge

In this section, let us introduce the entanglement swapping property of d-level Bell states and the notations we used in this work. For simplicity, symbols $\oplus$, $\ominus$ denote addition modulo d and subtraction modulo d, respectively, throughout the paper.

A d-level Bell basis contains states of the following form: [31]

$$|\psi\,(a,b)\rangle := \frac{1}{\sqrt{d}}\sum_{j=0}^{d-1}\zeta^{ja}\,|j\rangle|j\oplus b\rangle\,, \tag{1}$$

where $\zeta = e^{2\pi i/d}$, and a and b range from 0 to $d-1$. These states reduce to the familiar Bell states when $d=2$.

Suppose particles 1 and 2 and particles 3 and 4 are in states $|\psi(a_1,b_1)\rangle_{1,2}$ and $|\psi(a_2,b_2)\rangle_{3,4}$, respectively. Swapping particle 2 and 4 gives us [31]

$$\begin{aligned}&|\psi(a_1,b_1)\rangle_{1,2}\otimes|\psi(a_2,b_2)\rangle_{3,4}\\&=\frac{1}{d}\sum_{j,k=0}^{d-1}\zeta^{-jk}|\psi(a_1\oplus j,b_2\oplus k)\rangle_{1,4}\otimes|\psi(a_2\ominus j,b_1\ominus k)\rangle_{3,2}.\end{aligned} \tag{2}$$

We now consider the case where $n+1$ pairs of d-level Bell states are involved. Suppose these states are $|\psi(a_1,b_1)\rangle_{1,2}, |\psi(a_2,b_2)\rangle_{3,4},\ldots,|\psi(a_{n+1},b_{n+1})\rangle_{2n+1,2n+2}$. By repeatedly using Eq. (2), we can deduce the following

$$\begin{aligned}&|\psi(a_1,b_1)\rangle_{1,2}\otimes|\psi(a_2,b_2)\rangle_{3,4}\otimes\ldots\otimes|\psi(a_{n+1},b_{n+1})\rangle_{2n+1,2n+2}\\&=\frac{1}{d}\sum_{x_1,y_1=0}^{d-1}\zeta^{-x_1y_1}|\psi(a_1\oplus x_1,b_2\oplus y_1)\rangle_{1,4}\otimes|\psi(a_2\ominus x_1,b_1\ominus y_1)\rangle_{3,2}\\&\quad\otimes|\psi(a_3,b_3)\rangle_{5,6}\otimes\ldots\otimes|\psi(a_{n+1},b_{n+1})\rangle_{2n+1,2n+2}\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{d^2} \sum_{x_1, y_1=0}^{d-1} \sum_{x_2, y_2=0}^{d-1} \zeta^{-x_1 y_1 - x_2 y_2} |\psi(a_1 \oplus x_1, b_2 \oplus y_1)\rangle_{1,4} \\
&\quad \otimes |\psi(a_2 \ominus x_1 \oplus x_2, b_3 \oplus y_2)\rangle_{3,6} \\
&\quad \otimes |\psi(a_3 \ominus x_2, b_1 \ominus y_1 \ominus y_2)\rangle_{5,2} \\
&\quad \otimes |\psi(a_4, b_4)\rangle_{7,8} \otimes \ldots \otimes |\psi(a_{n+1}, b_{n+1})\rangle_{2n+1,2n+2} \\
&= \ldots \\
&= \frac{1}{d^{k-1}} \sum_{x_1, y_1=0}^{d-1} \sum_{x_2, y_2=0}^{d-1} \cdots \sum_{x_{k-1}, y_{k-1}=0}^{d-1} \zeta^{-\sum_{j=1}^{k-1} x_j y_j} |\psi(a_1 \oplus x_1, b_2 \oplus y_1)\rangle_{1,4} \\
&\quad \otimes |\psi(a_2 \ominus x_1 \oplus x_2, b_3 \oplus y_2)\rangle_{3,6} \\
&\quad \otimes |\psi(a_3 \ominus x_2 \oplus x_3, b_4 \oplus y_3)\rangle_{5,8} \\
&\quad \ldots \\
&\quad \otimes |\psi(a_k \ominus x_{k-1}, b_1 \ominus y_1 \ominus y_2 \ominus \ldots \ominus y_{k-1})\rangle_{2k-1,2} |\psi(a_{k+1}, b_{k+1})\rangle_{2k+1,2k+2} \\
&\quad \otimes \ldots \otimes |\psi(a_n, b_n)\rangle_{2n+1,2n+2} \\
&= \ldots \\
&= \frac{1}{d^n} \sum_{x_1, y_1=0}^{d-1} \sum_{x_2, y_2=0}^{d-1} \cdots \sum_{x_n, y_n=0}^{d-1} \zeta^{-\sum_{j=1}^{n} x_j y_j} |\psi(a_1 \oplus x_1, b_2 \oplus y_1)\rangle_{1,4} \\
&\quad \otimes |\psi(a_2 \ominus x_1 \oplus x_2, b_3 \oplus y_2)\rangle_{3,6} \\
&\quad \otimes |\psi(a_3 \ominus x_2 \oplus x_3, b_4 \oplus y_3)\rangle_{5,8} \\
&\quad \ldots \\
&\quad \otimes |\psi(a_k \ominus x_{k-1} \oplus x_k, b_{k+1} \oplus y_k)\rangle_{2k-1,2k+2} \\
&\quad \otimes \ldots \otimes |\psi(a_n \ominus x_n, b_1 \ominus y_1 \ominus y_2 \ominus \ldots \ominus y_n)\rangle_{2n+1,2}, \qquad (3)
\end{aligned}
$$

which can be depicted in Fig. 1.

If particles $(2j-1)$ and $(2j+2)$ $(j = 1, 2, \ldots, n)$ and particles $(2n+1)$ and 2 are measured in the d-level Bell basis and the measurement results are $|\psi(v_j, w_j)\rangle_{2j-1,2j+2}$ and $|\psi(v_{n+1}, w_{n+1})\rangle_{2n+1,2}$, respectively, then we have

$$\sum_{k=1}^{n+1} a_k \equiv \sum_{k=1}^{n+1} v_k \pmod{d}, \qquad (4)$$

$$\sum_{k=1}^{n+1} b_k \equiv \sum_{k=1}^{n+1} w_k \pmod{d}, \qquad (5)$$

according to Eq. (3). The measurement results will be used as keys to encrypt users' private inputs in our protocol.

The d-level computational basis is defined as the orthonormal set $\{|0\rangle, |1\rangle, \ldots, |d-1\rangle\}$. Performing the quantum Fourier transform on each element of this set then

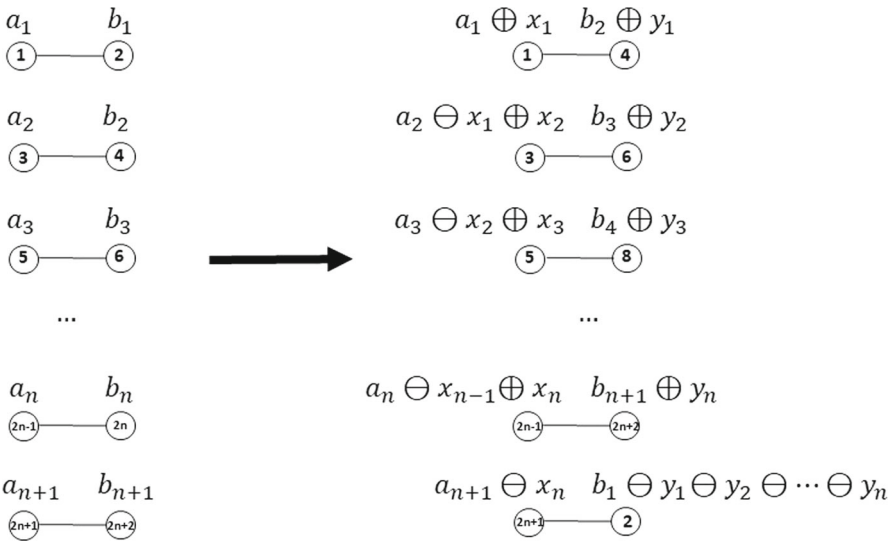


Fig. 1 Entanglement swapping of d-level Bell states

gives us

$$|f_k\rangle := F|k\rangle = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{kr} |r\rangle, \tag{6}$$

where $k = 0, 1, \ldots, d-1$, which forms another basis, namely the Fourier basis. States randomly chosen from the computational and the Fourier bases will be employed to detect the outside eavesdropping. In addition, $|j\rangle \in \{|0\rangle, |1\rangle, \ldots, |d-1\rangle\}$ can be expanded in the Fourier basis as:

$$|j\rangle = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{-jr} |f_r\rangle. \tag{7}$$

Equation (1) can thus be rewritten as:

$$\begin{aligned} |\psi\,(a_1, b_1)\rangle &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \zeta^{ja_1} |j\rangle |j \oplus b_1\rangle \\ &= \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \zeta^{ja_1} \frac{1}{\sqrt{d}} \sum_{m=0}^{d-1} \zeta^{-jm} |f_m\rangle \frac{1}{\sqrt{d}} \sum_{n=0}^{d-1} \zeta^{-(j\oplus b_1)n} |f_n\rangle \\ &= \frac{1}{\sqrt{d}} \cdot \frac{1}{d} \cdot \zeta^{b_1 n} \sum_{j,m,n=0}^{d-1} \zeta^{j(a_1-m-n)} |f_m\rangle |f_n\rangle \end{aligned}$$

$$= \frac{1}{\sqrt{d}} \cdot \zeta^{b_1 n} \sum_{\substack{m+n \\ \equiv a_1 (\bmod d)}} |f_m\rangle |f_n\rangle, \tag{8}$$

where the last equation follows from the fact $\sum_{j=0}^{d-1} \zeta^{jk} = d\delta_{k,0}$ and $\delta_{k,0} = d$ if $k = 0$, otherwise $\delta_{k,0} = 0$. Equation (8) will be utilized to detect the honesty of the malicious but non-collusive TP who is in charge of distributing d-level Bell states between users.

3 The proposed protocol

We are now ready to describe our protocol satisfying the following requirements:

(1) *Correctness* the modulo-d summation of all users' private inputs should be correct.
(2) *Security* an outside eavesdropper cannot obtain any information about each user's private input.
(3) *Privacy* no user can learn about other users' private inputs.
(4) *Fairness* All users get the summation result at the same time.

The classical and quantum channels used in our protocol are assumed to be authenticated and noiseless, respectively. The third party (TP) who distributes d-level Bell states between users and computes the modulo-d summation of all users' private inputs is assumed to be malicious but non-collusive; that is to say, TP is allowed to launch various attacks by using different quantum resources except the collusion with any dishonest user. For simplicity, TP will be donated by P_1 in the remainder of the paper.

Suppose there are n users $P_2, P_3, \ldots, P_{n+1}$, with P_i $(i = 2, 3, \ldots, n+1)$ having two private integers $m_{i1}, m_{i2} \in \{0, 1, \ldots, d-1\}$. P_1 helps them compute

$$\left(\sum_{i=2}^{n+1} m_{i1} \ (\text{mod } d), \sum_{i=2}^{n+1} m_{i2} \ (\text{mod } d) \right), \tag{9}$$

without learning any information about their private integers.

Our protocol works as follows:

(1) *Entanglement distribution* P_1 generates $(n+1)(1+R)$ ordered d-level Bell states, each of which is randomly selected from the d-level Bell basis,

$$K_i = \left(|\psi(a_i^1, b_i^1)\rangle_{2i-1,2i}, |\psi(a_i^2, b_i^2)\rangle_{2i-1,2i}, \ldots, |\psi(a_i^{1+R}, b_i^{1+R})\rangle_{2i-1,2i} \right), \tag{10}$$

where $i = 1, 2, \ldots, n+1$. P_1 then picks up the first component of each d-level Bell state from K_i to form an ordered sequence S^i_{2i-1}, leaving the second component of the states forming another ordered sequence S^i_{2i+2}. P_1 then prepares n sets of decoy states. Each set contains d decoy states that are randomly chosen from the computational basis or from the Fourier basis. Afterward, P_1 inserts randomly each set of decoy states into S^j_{2j-1} and $S^{j+1}_{(2j+2) \bmod (2n+2)}$, $(j = 2, 3, \ldots, n+1)$,

forming new sequences $(S^j_{2j-1})'$ and $(S^{j+1}_{(2j+2) \bmod (2n+2)})'$, respectively. At last, P_1 sends $(S^j_{2j-1})'$ and $(S^{j+1}_{(2j+2) \bmod (2n+2)})'$ to P_j and keeps S^1_1 and S^2_4 in his hands. Note that, at this stage, the positions of the decoy states in each sequence and the information about all the d-level Bell states are only known to P_1.

(2) *Detection for outside attacks* After confirming that P_j $(j = 2, 3, \ldots, n+1)$ has received $(S^j_{2j-1})'$ and $(S^{j+1}_{(2j+2) \bmod (2n+2)})'$ $(j = 2, 3, \ldots, n+1)$ from P_1, P_1 and P_j check whether the transmission between them is secure against outside attacks. Concretely, P_1 first announces to P_j the position and the corresponding bases of the decoy states. Based on the announced information, P_j measures these decoy states in the correct bases and publishes at random half the measurement results. Next, P_1 announces the original states of the other half of the decoy states. They then compute the error rate by comparing the measurement results with their corresponding original states. If the error rate exceeds a certain threshold, the protocol will abort and restart from Step (1); otherwise, it will continue to the next step.

(3) *Detection for the honesty of* P_1 By removing the d decoy states from $(S^j_{2j-1})'$ and $(S^{j+1}_{(2j+2) \bmod (2n+2)})'$ $(j = 2, 3, \ldots, n+1)$, P_j obtains S^j_{2j-1} and $S^{j+1}_{(2j+2) \bmod (2n+2)}$ $(j = 2, 3, \ldots, n+1)$. All the n users then detect whether the genuine d-level Bell states are shared in an honest way. After Step (1), (1+R) d-level Bell states $|\psi(a^k_1, b^k_1)\rangle_{1,2}$ $(k = 1, 2, \ldots, 1+R)$ should be shared between P_1 and P_{n+1}; (1+R) d-level Bell states $|\psi(a^k_j, b^k_j)\rangle_{2j-1,2j}$ $(k = 1, 2, \ldots, 1+R, j = 2, 3, \ldots, n+1)$ should be shared between P_j and P_{j-1}.

(3.1) To examine the genuineness of the d-level Bell states shared between P_1 and P_{n+1}, P_{n+1} first randomly selects R d-level Bell states between them and requests P_1 to announce the corresponding original states. Next, P_{n+1} randomly measures each corresponding component in S^1_2 in the computational basis or in the Fourier basis. These measurement results are only known to himself. Later on, P_{n+1} notifies P_1 to measure the corresponding components in S^1_1 in the same basis as P_{n+1} does and announce the measurement results. According to Eq. (8), P_{n+1} examines if these measurement results are correlated with each other, and computes the error rate. For example, if the original state announced by P_1 is $|\psi(a_1, b_1)\rangle_{1,2}$, and the measurement results in the computational basis are r_1 and r_2, then the equation $r_2 - r_1 \equiv b_1 (\bmod\ d)$ should be obtained; otherwise, the equation $r_1 + r_2 \equiv a_1 (\bmod\ d)$ should hold. If the error rate is greater than a certain threshold, the protocol will abort and restart from Step (1). Otherwise, the protocol will continue.

(3.2) To examine the genuineness of the d-level Bell states shared between P_2 and P_1, P_2 also employs R d-level Bell states to complete this detection by using the similar method as that used by P_{n+1}. If the error rate is greater than the threshold, the protocol will abort and restart from Step (1). Otherwise, the protocol will continue.

(3.3) To examine the genuineness of the d-level Bell states shared between P_j and P_{j-1} $(j = 3, 4, \ldots, n+1)$, P_j randomly chooses $R/2$ d-level Bell states between them and asks P_1 to announce the corresponding original states. P_j

then measures each corresponding component in S^j_{2j-1} randomly in the computational basis or in the Fourier basis, publishing the measurement results. Afterward, P_{j-1} measures the corresponding components in S^j_{2j} in the same basis, publicizing his measurement results. P_j and P_{j-1} can then check if these measurement results are correlated based on Eq. (8). P_{j-1} utilizes the same procedure as P_j dose with another $R/2$ randomly chosen Bell states to derive the relation of P_j's and P_{j-1}'s measurement results. If the error rate in either case is greater than the threshold, the protocol will abort and restart from Step (1). Otherwise, they have the guarantee that the states shared between them are genuine Bell states and distributed in an honest way, and the protocol will continue.

(4) *Private input encryption* P_i $(i = 1, 2, \ldots, n+1)$ discards R d-level Bell states used for P_1's honesty detection, resulting in only $n+1$ d-level Bell states shared between every two users. Suppose these $n+1$ states are $|\psi(a_1, b_1)\rangle_{1,2}$ shared between P_1 and P_{n+1} and $|\psi(a_j, b_j)\rangle_{2j-1,2j}$ $(j = 2, 3, \ldots, n+1)$ shared between P_j and P_{j-1}. Note that only P_1 has the knowledge of (a_j, b_j). P_1 measures particles 1 and 4 in the d-level Bell basis, obtaining the measurement result $|\psi(v_1, w_1)\rangle_{1,4}$. P_j $(j = 2, 3, \ldots, n+1)$ also measures particles $(2j-1)$ and $((2i+2) \bmod (2n+2))$ in the d-level Bell basis and gets the measurement results $|\psi(v_j, w_j)\rangle_{2j-1,(2j+2) \bmod (2n+2)}$. P_j then encrypts his private inputs (m_{j1}, m_{j2}) by computing

$$c_{j1} \equiv m_{j1} \oplus v_j, \tag{11}$$

$$c_{j2} \equiv m_{j2} \oplus w_j, \tag{12}$$

which are later sent to P_1 through authenticated classical channels.

(5) *Summation.* After receiving all (c_{j1}, c_{j2}) $(j = 2, 3, \ldots, n+1)$, P_1 computes

$$t_1 \equiv \left(v_1 \oplus \sum_{j=2}^{n+1} c_{j1} \right) - \left(\sum_{j=1}^{n+1} a_i \right) \pmod{d}, \tag{13}$$

$$t_2 \equiv \left(w_1 \oplus \sum_{j=2}^{n+1} c_{j2} \right) - \left(\sum_{j=1}^{n+1} b_i \right) \pmod{d}, \tag{14}$$

and announces (t_1, t_2) to $P_2, P_3, \ldots, P_{n+1}$.

4 Analysis

Let us now analyse the correctness of our protocol in Sect. 4.1 as well as its security in Sect. 4.2.

4.1 Correctness analysis

We begin with the correctness analysis. We show here that if all users honestly provide their private inputs, they can eventually obtain the correct summation result

$$\left(\sum_{i=2}^{n+1} m_{i1} \pmod d, \sum_{i=2}^{n+1} m_{i2} \pmod d\right). \tag{15}$$

In Step (4) of the proposed protocol, $P_j (j = 2, 3, \ldots, n+1)$ uses his measurement result (v_j, w_j) as a key to encrypt his private inputs (m_{j1}, m_{j2}) by Eqs. (11) and (12) to generate ciphertexts (c_{j1}, c_{j2}) that are then sent to P_1 through authenticated classical channels.

In Step (5), by employing Eqs. (4) and (5), P_1 calculates and announces the following

$$\begin{aligned}
t_1 &\equiv \left(v_1 \oplus \sum_{j=2}^{n+1} c_{j1}\right) - \left(\sum_{j=1}^{n+1} a_i\right) \pmod d \\
&\equiv \left(v_1 \oplus \sum_{j=2}^{n+1} (m_{j1} \oplus v_i)\right) - \left(\sum_{j=1}^{n+1} a_i\right) \pmod d \\
&\equiv \left(\sum_{j=1}^{n+1} v_j \oplus \sum_{j=2}^{n+1} m_{j1}\right) - \left(\sum_{j=1}^{n+1} a_i\right) \pmod d \\
&\equiv \sum_{j=2}^{n+1} m_{j1} \pmod d, \qquad (16) \\
t_2 &\equiv \left(w_1 \oplus \sum_{j=2}^{n+1} c_{j2}\right) - \left(\sum_{j=1}^{n+1} b_i\right) \pmod d \\
&\equiv \left(w_1 \oplus \sum_{j=2}^{n+1} (m_{j2} \oplus w_i)\right) - \left(\sum_{j=1}^{n+1} b_i\right) \pmod d \\
&\equiv \left(\sum_{j=1}^{n+1} w_j \oplus \sum_{j=2}^{n+1} m_{j2}\right) - \left(\sum_{j=1}^{n+1} b_i\right) \pmod d \\
&\equiv \sum_{j=2}^{n+1} m_{j2} \pmod d, \qquad (17)
\end{aligned}$$

which are the modulo-d summation result of n users' private inputs. And our protocol thus fulfils the correctness requirement.

4.2 Security analysis

Let us now move on to the security analysis. We have to show that the proposed protocol can resist both outside and participant attacks, and it thus satisfies the security, privacy and fairness requirements mentioned in Sect. 3. Outside attacks may potentially affect our protocol via the initial entanglement distribution. Note that P_1, as a special participant, cannot collude with any dishonest user, but he can launch attacks within boundaries of quantum mechanics. Other dishonest users may collude to steal the rest users' private inputs. We will show how we can detect these attacks by using decoy states and extra d-level Bell states. Trojan horse attacks [28,32,33], such as the delay-photon Trojan horse attacks as well as the invisible photon eavesdropping Trojan horse attack, usually exist in quantum communication protocols where quantum states are encoded and relayed. As the entanglement distribution occurs only once in a one-way communication, these attacks become invalid to our protocol.

4.2.1 Analysis of outside attacks

We now give a detailed analysis of the security of our protocol against outside attacks. The decoy-state technique is utilized in our protocol. The idea comes from the first quantum key distribution protocol by Bennett and Brassard [34] that has been proven to be unconditional secure [35]. Our protocol can thus resist a variety of outside attacks, such as the intercept-resend attack, the measurement-resend attack, the entanglement-measurement attack, and the denial-of-service (DOS) attack. Without loss of generality, the most general strategy that the outside attacker, Eve, can adopt is that she uses the operation U_E to entangle the quantum state transmitted from P_1 to P_j with her ancillary quantum system $|E\rangle$, which can be characterized as

$$U_E|j\rangle|E\rangle = \sum_{i=0}^{d-1} a_{ji}|i\rangle|e_{ji}\rangle, \tag{18}$$

where $|j\rangle \in \{|0\rangle, |1\rangle, \ldots, |d-1\rangle\}$ and $\sum_{i=0}^{d-1} |a_{ji}|^2 = 1$. We show that any attempt to eavesdrop by Eve will inevitably change the photon statistics and expose her.

On the one hand, suppose the decoy state comes from the computational basis, and it is measured in this basis. If Eve introduces no error, the following equation

$$a_{ij} = \begin{cases} 0, & i \neq j, \\ 1, & i = j, \end{cases} \tag{19}$$

should be satisfied according to Eq. (18).

On the other hand, by using Eqs. (7) and (19), the states from the Fourier basis performed by the operation U_E become

$$U_E|f_j\rangle|E\rangle = U_E F|j\rangle|E\rangle = U_E \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{jr}|r\rangle|E\rangle$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{jr} U_E |r\rangle |E\rangle = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{jr} \sum_{i}^{d-1} a_{ri} |i\rangle |e_{ri}\rangle \\
&= \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \sum_{i=0}^{d-1} \zeta^{jr} a_{ri} |i\rangle |e_{ri}\rangle = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{jr} a_{rr} |r\rangle |e_{rr}\rangle \\
&= \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} \zeta^{jr} \frac{1}{\sqrt{d}} \sum_{p=0}^{d-1} \zeta^{-rp} |f_p\rangle |e_{rr}\rangle \\
&= \frac{1}{d} \sum_{r=0}^{d-1} \sum_{p=0}^{d-1} \zeta^{jr-pr} |f_p\rangle |e_{rr}\rangle \\
&= \frac{1}{d} \sum_{p \neq j} \sum_{r=0}^{d-1} \zeta^{jr-pr} |f_p\rangle |e_{rr}\rangle + \frac{1}{d} \sum_{r=0}^{d-1} |f_j\rangle |e_{rr}\rangle. \qquad (20)
\end{aligned}
$$

Thus, if the decoy state is from the Fourier basis and measured in this basis, in order to introduce no error by Eve, the following condition

$$
\frac{1}{d} \sum_{p \neq j} \sum_{r=0}^{d-1} \zeta^{jr-pr} |f_p\rangle |e_{rr}\rangle = 0, \qquad (21)
$$

should hold, where the 0 on the right side represents zero vector, and j ranges from 0 to $d-1$, which leads to

$$
|e_{00}\rangle = |e_{11}\rangle = \cdots = |e_{(d-1)(d-1)}\rangle, \qquad (22)
$$

that is to say, Eve does not introduce any error only when her ancillary quantum state and the decoy state are product states. Therefore, Eve can learn nothing about users' private inputs.

4.2.2 Analysis of participant attacks

Let us now study the security of our protocol against participant attacks. We need to account for two kinds of attacks: one from P_1 who distributes the d-level Bell states between users, and the other from dishonest users.

Analysis of Attacks from P_1. We first analyse the attacks from P_1, who cannot collude with any dishonest user. Without loss generality, suppose that P_1 attempts to obtain P_k' private inputs (m_{k1}, m_{k2}) $(k \neq 2, n+1)$. To achieve this, P_1 may carry out the entanglement swapping attack as follows: P_1 first distributes two d-level Bell states shared between him and P_k. P_1 can then derive P_k's private key (v_k, w_k) using Eq. (1), because he knows the initial d-level states and his own measurement result in the d-level Bell basis; P_1 can thus obtain P_k's private inputs. Another attack is that P_1 distributes a d-level Bell state to P_k without sharing it with other user. It is easy for P_1 to obtain P_k's measurement result, and, therefore, to get P_k's private inputs.

These attacks, however, will be detected in Step (3) where the genuineness of the d-level Bell states shared between P_j and P_{j-1} (between P_{j+1} and P_j) is examined. In what follows, we show that the entanglement swapping attack is invalid to our protocol. Other attacks from P_1 can be analogously analysed as the security relies on the genuine d-level states shared between users in the honest way.

As we can see from Eq. (8) in Sect. 2, if a d-level Bell state shared between P_k and P_{k-1} is known to them and each component of the state is measured in the same basis (both in the computation basis or both in the Fourier basis), then their measurement results should be correlated. For example, if the state is $|\psi(a,b)\rangle$ and their measurement results in the computational basis are r_k and r_{k+1}, respectively. They then ought to have $r_{k+1} - r_k \equiv b \pmod d$. Otherwise, they should attain $r_{k+1} + r_k \equiv a \pmod d$. To obtain the maximum probability that P_1 passes the detection, he can distribute only one fake d-level Bell state between P_k and P_{k-1} and another fake one between P_{k+1} and P_k, and then share two genuine d Bell states between him and P_k, and P_{k-1}, respectively. He can then start the entanglement swapping attack. In this case, these two fake states cannot be used for detection. The probability of passing the detection thus become $1/(1+R)^2$. For the cases where $k=2$ or $k=n+1$, this probability turns into $1/(1+R)$ due to the fact that P_1 can only pass the detection between him and P_k. These two probabilities will close to 1 when R is large enough. As a consequence, P_1 is unable to take users' private inputs.

Analysis of Attacks from dishonest users

In this part, we will show that $n-2$ dishonest users will even fail to obtain just only one of the other two honest users' private inputs. Suppose that these dishonest users intend to achieve P_p's ($p \neq 1$) private inputs.

For the cases where $p=2$ or $p=n+1$, as one component of the d Bell state belongs to P_1 who cannot collude with dishonest users, they are unable to obtain useful measurement results in the d-level Bell basis. They can thus learn nothing about P_2's and P_{n+1}'s private inputs.

We now consider the cases where $p \neq 2$ and $p \neq n+1$. If the other honest user P_q is not P_p's neighbour, namely $q \neq p+1$ and $q \neq p-1$. Dishonest users P_{p-1} and P_{p+1} may initiate entanglement swapping attack as each of them has one component that is part of the d-level Bell state shared between P_{p-1} and P_p (between P_{p+1} and P_p). Suppose the state shared between P_p and P_{p-1} is $|\psi(a_p,b_p)\rangle_{2p-1,2p}$ and the state shared between P_{p+1} and P_p is $|\psi(a_{p+1},b_{p+1})\rangle_{2p+1,2p+2}$. We then have

$$\begin{aligned}
&|\psi(a_p,b_p)\rangle_{2p-1,2p} \otimes |\psi(a_{p+1},b_{p+1})\rangle_{2p+1,2p+2} \\
&= \frac{1}{d}\sum_{j,k=0}^{d-1} \zeta^{-jk} |\psi(a_p \oplus j, b_{p+1} \oplus k)\rangle_{2p-1,2p+2} \\
&\quad \otimes |\psi(a_{p+1} \ominus j, b_p \ominus k)\rangle_{2p+1,2p},
\end{aligned} \tag{23}$$

where particles $(2p-1)$ and $(2p+2)$ belong to P_p, particle $(2p)$ belongs to P_{p-1}, and particle $(2p+1)$ is in P_{p+1}'s hands. To derive P_p's measurement result in the d-level Bell basis, P_{p-1} and P_{p+1} need to get the information about the shared d-level Bell states, which is impossible as it is only known to P_1. They thus fail to learn about P_p's

private inputs. If the other honest user P_q is P_p's, namely $q = p + 1$ or $q = p - 1$. The analysis is similar. In this case, they cannot even attain the modulo-d summation of P_p' and P_q's private integers. Let alone P_p's private inputs.

Note that any $(n - 1)$ dishonest users can easily derive the rest one's private inputs when the summation result is revealed. Our protocol is, therefore, secure against $(n - 2)$-party collusive attacks.

5 Comparison and discussion

In this section, we first compare our protocol with existing quantum modulo-d summation protocols in terms of qudit cost, quantum resource, and quantum operations. And then we discuss how to replace the cat states in the protocols of Ref. [30] with the d-level Bell states, which facilitates the protocol's implementation.

The qudit cost of a quantum summation protocol is defined as the number of qudits (including the qudits used for security detection) required to obtain a modulo-d summation of n private dits, where n is the number of users.

For simplicity, we assume that all protocols use n decoy qudits to check eavesdropping whenever necessary.

Shi et al. [22] utilized qubits to construct modulo-d summation protocols, in which $d = 2^m$ and m is the number of qubits. For comparison, we regard d qubits as a qudit. As the authors mentioned, their protocol I can only protect the private input of the initiator, who starts the protocol. To overcome this weakness, they proposed protocol III based on protocol I, which can be secure against attacks from any less $n - 1$ dishonest users. To compute the modulo-d summation of n private dits provided by n users, protocol III should call protocol I $n + 1$ times, with each time, protocol I consuming $n + 1$ qudits. Totally, protocol III costs $(n + 1)(n + 1) = n^2 + 2n + 1$ qudits. Their protocol needs d-level $(n + 1)$-partite entangled d-level states and quantum operations, including the (inverse) quantum Fourier transform, the controlled-NOT (CNOT) operation, an oracle operation, and single qubit measurement. Protocols II and IV in their work are about modulo-d multiplication, which are not discussed here.

Yang et al. [36] presented a secure multi-party summation protocol based on d-level n-partite entangled states, where n is the number of users. A d-level n-partite entangled state is required to obtain a modulo-d summation of n private dits. Besides, $n(n - 1)$ decoy qudits are needed to prevent outside attacks, and $(n - 1)$ d-level n-partite entangled states should be employed to detect the initiator's honesty. The qudit cost thus becomes $n + n(n - 1) + n(n - 1) = 2n^2 - n$. In their protocol, the quantum Fourier transform and unitary operations related to users' private inputs are necessary. The single qudit measurements are performed to read the summation result.

In our previous work [37], d-level single quantum systems are employed to design a multi-party quantum summation protocol, where d is limited to a prime integer. Only one d-level single state is needed to compute a dit sum. However, it needs relaying $n + 1$ times and each time, n decoy qudits are required to detect eavesdropping. The qudit cost is thus $1 + (n + 1)n = n^2 + n + 1$. The protocol needs two unitary operations to combine to transform an element in a d-level basis to another element in another

basis. Besides, each participant should be equipped with quantum state generators and the measurement devices in $d+1$ different bases.

Ji et al.'s [30] protocol I is probabilistic can easily be modified into a modulo-d summation protocol. The only change is the last step, where the participant, who helps the computation, can just perform the summation in modulo d instead of the general one. The protocol then becomes deterministic. In this case, their protocol I needs a d-level $(n+1)$-partite cat state and n d-level Bell states. Besides, TP and n users have to transmit particles to each other, which requires $2n^2$ qudits for eavesdropping detection. Hence, their protocol's qudit cost is $(n+1)+2n+2n^2 = 2n^2+3n+1$. In addition, their protocols need the measurement in the n-partite cat basis, which is not available when n is very large with current quantum technology. Their protocol II based on protocol I aims at solving other problems, which is omitted here.

In our protocol in this work, we take advantage of the entanglement swapping property of d-level Bell states to devise a quantum modulo-d summation protocol, where a two-dit summation of n users' private inputs, each with two dits, is computed. $n+1$ pairs of d-level Bell states are required for the computation, and n^2 decoy qudits are used for outside eavesdropping detection. Also, n^2 pairs d-level Bell states are utilized to check the initiator's honesty. The qudit cost is, therefore, $((n+1)+2n+n^2+2n^2)/2 = (3n^2+3n+1)/2$. The quantum operations users need include only the d-level Bell measurement and the single qudit measurement.

We can see, from Table 1, that except the protocol in Ref. [37], other protocols require entangled states. It seems that the protocol in Ref. [37] is the best one because

Table 1 Comparison between quantum modulo-d summation protocols in terms of their qudit cost, quantum resources, and quantum operations

Protocols	Qudit cost	Quantum resource	Quantum operations
Protocol III of Ref. [22]	n^2+2n+1	d-level n-partite entangled state	Quantum Fourier transform, CONT operation, oracle operation, inverse quantum Fourier transform, single qubit measurement
Protocol of Ref. [36]	$2n^2-n$	d-level n-partite entangled state, d-level single decoy state	Quantum Fourier transform, unitary operation U_k, single qudit measurement
Protocol of Ref. [37]	n^2+n+1	d-level single encoding state and d-level single decoy state	Two unitary operations (X_d and Y_d), single qudit measurement
Protocol I of Ref. [30]	$2n^2+3n+1$	d-level $(n+1)$-partite entangled state, e-level bipartite state and d-level single decoy state	A unitary operation ($U_{u,v}$), single qudit measurement, d-level Bell measurement, $(n+1)$-partite cat state measurement
Our Protocol	$(3n^2+3n+1)/2$	d-level bipartite entangled state and d-level single decoy state	Single qudit measurement, d-level Bell measurement

of its lowest qudit cost and fewest quantum operations. However, every user needs quantum generators to generate decoy qudits, as well as quantum measurement devices in $d+1$ different bases. Moreover, this protocol cannot be modified for the applications in quantum anonymous ranking and voting. In contrast, our protocol based on d-level bipartite states can be modified for these applications as the protocols do in Ref. [30]. Further, users in our protocol only need single qudit measurements and d-level measurements, which is much more available with current technology.

We are now interested in how to replace the cat states with d-level Bell states in the protocols in Ref. [30] to make the protocols simple. We mainly focus on the protocol I as it is the primitive for their protocol II.

Let us first present their idea. For simplicity, we consider the case where each user has one private integer. In the protocol I, there are n users $P_1, P_2, \ldots, P_n$ and P_i has a private input $x_i \in \{0, 1, \ldots, d-1\}$. They wish to compute the summation $\sum_{i=1}^{n} x_i$ instead of $\sum_{i=1}^{n} x_i \pmod d$. To finish this task, they use the entanglement swapping property between the cat state and the Bell state as follows:

$$\begin{aligned} &|\psi\left(u_1, u_2, \ldots, u_n\right)\rangle_{1,2,\ldots,n} \otimes \left|\psi\left(v, v'\right)\right\rangle_{s,s'} \\ &= \frac{1}{d} \sum_{k,l}^{d-1} \zeta^{kl} \left|\psi\left(u_1 \oplus k, u_2, u_3, \ldots, v' \oplus l, \ldots, u_n\right)\right\rangle_{1,2,\ldots,s',\ldots,n} \\ &\quad \otimes |\psi\left(v \ominus k, u_m \ominus l\right)\rangle_{s,m}. \end{aligned} \tag{24}$$

The process can be described in the following:

(1) TP prepares a cat state $|\psi\left(u_0, u_1, \ldots, u_n\right)\rangle$ and P_i $(i = 1, 2, \ldots, n)$ generates a d-level Bell state $\left|\psi\left(v_i, v_i'\right)\right\rangle$, where $v_i' = x_i$. TP then tells P_i the mark u_i, as well as sending him the particle with the mark u_i by using decoy-state detection technique.

(2) After receiving the particle with the mark u_i $(i = 1, 2, \ldots, n)$ from TP, P_i measure it and the particle with the mark v_i in the d-level Bell basis, which make the entanglement swapping happen; see Fig. 2 [30] for clarity.

(3) Using u_i, all parties collaborate to compute

$$S_P = \sum_{i=1}^{n} l_i, \tag{25}$$

which is then announced to TP.

(4) P_i $(i = 1, 2, \ldots,)$ sends the other component that is not measured to TP by decoy-state detection technique, and TP measures his cat state, obtaining the marks

$$\left(u_0 + \sum_{i=1}^{n} k_i\right) \bmod d, v_1' \oplus l_1, v_2' \oplus l_2^j, \ldots, v_n' \oplus l_n. \tag{26}$$

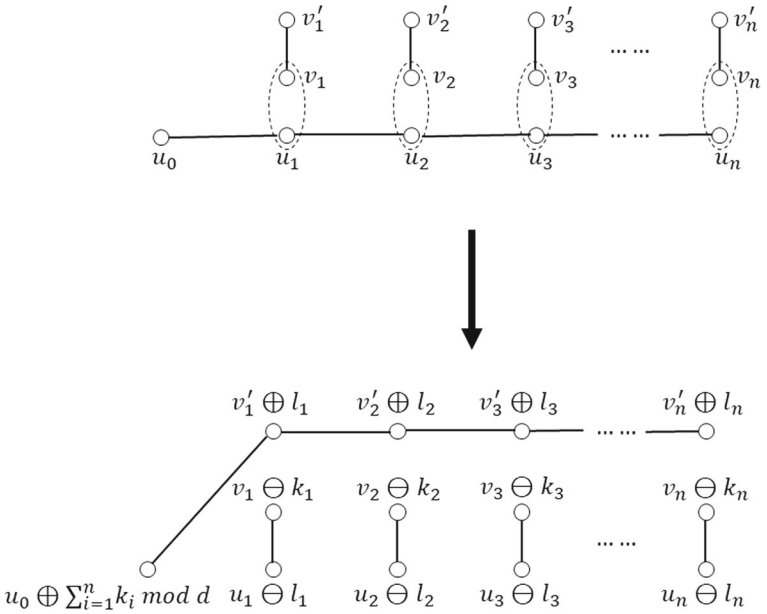


Fig. 2 The graphical description of entanglement swapping between a cat state and n Bell states

TP then computes

$$\sum_{i=1}^{n} \left(v_i' \oplus l_i\right) - S_P = \sum_{i=1}^{n} v_i', \tag{27}$$

which is the summation $\sum_{i=1}^{n} x_i$.

In what follows, we show that the cat state can be replaced with n d-level Bell states.

In Step (1), instead of preparing a cat state, TP generates n d-level Bell states $|\psi(u_i', u_i)\rangle$ $(i = 1, 2, \ldots, n)$, and P_i $(i = 1, 2, \ldots, n)$ generates a Bell state $\left|\psi\left(v_i, v_i'\right)\right\rangle$, where $v_i' = x_i$. TP then announces to P_i the mark u_i as well as sending him the particle with the mark u_i by using decoy-state detection technique.

Steps (2) and (3) remain the same. But note that, Step (2) makes the entanglement swapping happen; see Fig. 3.

In step (4), after receiving the other component not measured from P_i, TP measures his n Bell states in the d-level Bell basis and obtains the marks

$$(u_i' \oplus k_i, v_i' \oplus l_i). \tag{28}$$

TP is then able to compute TP then computes

$$\sum_{i=1}^{n} \left(v_i' \oplus l_i\right) - S_P = \sum_{i=1}^{n} v_i', \tag{29}$$

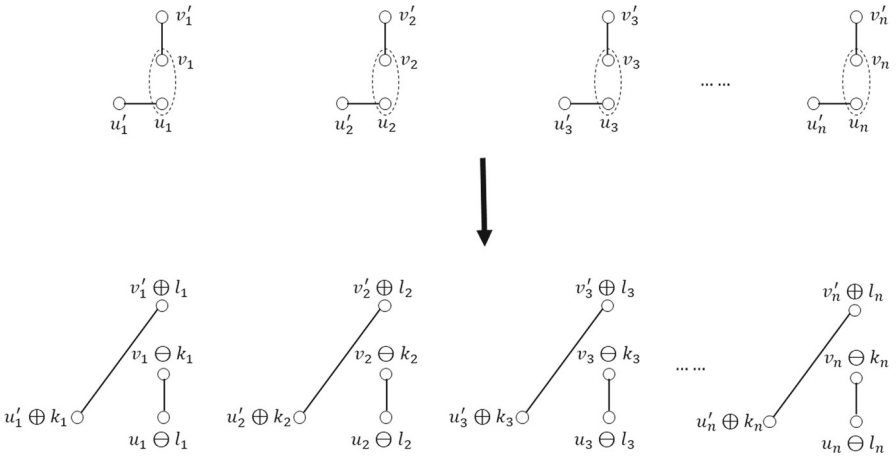


Fig. 3 The graphical description of entanglement swapping between Bell states

which is the summation $\sum_{i=1}^{n} x_i$.

For the values of $l_1, l_1, \ldots, l_n$ obtained from the methods in Figs. 2 and 3, respectively, they are equivalent, which means that a cat state can be replaced with n d-level Bell states. Thus, the replaced version has the same successful probability that TP has a correct result as the protocol I in Ref. [30].

For the security, as pointed out by Ref. [38], Ji et al.'s protocols become insecure if $l_i (i = 1, 2, \ldots, n)$ were sent to TP via the public channel. Thus, our replaced version also suffers such a security loophole. This security loophole can be closed by the method proposed in Ref. [38], namely $P_1, P_2, \ldots, P_n$ pre-share a private key sequence.

Based on that, we now consider the most powerful attack, where $n - 2$ dishonest users desire to collude to steal the rest two users' private inputs. Suppose these dishonest users are $P_1, P_2, \ldots, P_{i-1}, P_{i+1}, \ldots, P_{j-1}, P_{j+1}, \ldots, P_n$, and they intend to collude to obtain P_i's private input, x_i, and P_j's private input, x_j. Although they are able to learn about the value of $l_i + l_j$ from S_P and their l_m $(m = 1, 2, \ldots, i-1, i+1, \ldots, j-1, j+1, \ldots, n)$ in accordance with Eq. (25), they cannot obtain the values of v_i' and v_j' as they know nothing about $v_i' \oplus l_i$ and $v_j' \oplus l_j$. As other quantum summation protocols, any $n - 1$ dishonest users can get the rest one's private input once the summation is revealed.

6 Conclusions

We first designed a multi-party quantum modulo-d summation by taking advantage of the entanglement swapping property of d-level Bell states. A malicious but non-collusive third party (TP), also known as P_1 in our protocol, plays a role of distributing entanglement shared between users in such a way that he can finally obtain the summation result according to n users' cipher-texts. The correctness and the security of the proposed protocol were analysed in detail. Afterward, we gave a comparison of

different quantum summation in modulo-d protocols. Finally, we discussed how the cat state in Ji et al. [30] protocols can be replaced with n d-level Bell states to solve the same problems.

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